

COLLISION-MARKED TRAJECTORY CLUSTERS AND JOINT DYNAMIC LARGE DEVIATIONS FOR DETERMINISTIC HARD SPHERES

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ABSTRACT. We construct a compatible measure-valued kinetic normal-trace graph and its positive boundary-renewal hierarchy. Recollisions are controlled in the actual chronological variables by a flux-weighted exterior-power estimate for the physical hard-sphere Jacobi cocycle, followed by factorial collision-tree summation. A source-uniform scaled block theorem and an exactly conservative finite-cell recovery yield the good grand-canonical and microcanonical density–actual-contact LDPs.

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1. ROUND-NINE RECONSTRUCTION: COMPATIBLE KINETIC TRACES, PHYSICAL JACOBI CONTROL, AND SOURCE-UNIFORM SEWING

The boundary variable is not an arbitrary measure independent of the interior state. It is the normal trace of a divergence-measure transport state and is part of a closed Green graph. Recollision estimates are performed in the actual chronological variables. Coordinate resets are not used to erase small singular values; instead an exterior-power estimate for the physical hard-sphere Jacobi cocycle is integrated with the incoming flux measure and summed together with the factorial collision-tree weights.

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1.1. The compatible normal-trace graph. For $k \geq 1$ let

$$\Omega_k^\varepsilon = \{Z_k = (x_i, v_i)_{i \leq k} : |x_i - x_j| > \varepsilon\}$$

and let Γ_{ij}^- be its incoming binary face. A level state is a positive finite measure g_k on Ω_k^ε such that the transport current

$$\mathcal{J}_k = (g_k, v_1 g_k, \dots, v_k g_k)$$

is a divergence-measure field with finite weighted divergence. Its incoming normal trace is denoted $\gamma_{k,ij}^-$; it is allowed to have atoms. Put

$$\mathfrak{G}_k^\beta = \{(g_k, \gamma_k^-) : \gamma_k^- = \text{Tr}^- \mathcal{J}_k, \|g_k\|_{W_\beta} + \|\text{div} \mathcal{J}_k\|_{W_\beta} + \|\gamma_k^-\|_{W_\beta} < \infty\}.$$

The norm includes the outgoing trace obtained by specular reflection and the corner compatibility relations in the sense of normal currents.

Theorem 1.1 (Closed Green graph and reflected transport). *The graph $\mathfrak{G}_k^\beta$ is closed. For every bounded C^1 test ϕ with the velocity weight,*

$$\int_{\Omega_k^\varepsilon} (\partial_t + V_k \cdot \nabla_X) \phi dg_k + \int_{\Omega_k^\varepsilon} \phi d(\text{div} \mathcal{J}_k) = \sum_{i < j} \int_{\Gamma_{ij}^-} \phi (V_k \cdot n_{ij}) d\gamma_{k,ij}^-.$$

Specular reflection identifies the incoming and outgoing traces and defines a positive mass-preserving transport semigroup $U_k^\varepsilon(t)$ on the closed graph. Boundary Dirac masses are included; no L^1 approximation of a boundary atom is required.

Proof. Approximate the hard-sphere domain by deleting a grazing/corner neighbourhood and apply the Gauss–Green theorem for divergence-measure fields. The normal trace is continuous for strict convergence of the current and its divergence. Uniform weighted flux bounds let the deleted neighbourhood shrink; the corner sets have codimension at least two in flux measure. This proves the Green identity and closedness. Free streaming followed by the measure-preserving specular involution gives a positive isometry on smooth graph states. Its graph closure is strongly continuous and mass preserving. The construction acts directly on finite normal traces and therefore retains atoms. $\square$

Let $B_{k,k+1}$ be the creation/contact operator defined by integrating the incoming trace of the $(k+1)$ st particle against the physical collision flux. With factorial hierarchy norm

$$\|G\|_{\alpha,\beta} = \sum_{k \geq 0} \frac{e^{\alpha k}}{k!} \|(g_k, \gamma_k^-)\|_{\mathfrak{G}_k^\beta},$$

write $\mathcal{U}^\varepsilon(t)$ for the Dyson–Phillips boundary-renewal series.

Proposition 1.2 (Positive boundary-renewal hierarchy). *For $\alpha' > 0$ smaller than α and T below the declared kinetic horizon,*

$$\mathcal{U}^\varepsilon(t) = \sum_{n \geq 0} \int_{0 < t_1 < \dots < t_n < t} U(t - t_n) B \cdots B U(t_1) dt_1 \cdots dt_n$$

converges from $\mathfrak{X}_{\alpha,\beta}$ to $\mathfrak{X}_{\alpha',\beta'}$. It is positive, satisfies the hierarchy Green identities, and obeys

$$\|\mathcal{U}^\varepsilon(t)G\|_{\alpha',\beta'} \leq C_T \|G\|_{\alpha,\beta}.$$

Proof. The coarea definition of $B_{k,k+1}$ and Theorem 1.1 give

$$\|B_{k,k+1}G_{k+1}\|_{\mathfrak{G}_k^{\beta'}} \leq C(k+1)\|G_{k+1}\|_{\mathfrak{G}_{k+1}^\beta}.$$

The loss from grazing is paid by one velocity weight and by the incoming flux already present in the graph norm. Multiplication by $(k+1)$ is absorbed by the factorial label weight. The ordered time simplex supplies $t^n/n!$; summing proves the estimate and positivity. Applying the Green identity at every term proves compatibility of the renewed traces. $\square$

1.2. The physical youngest-fork variables. Fix a chronological collision tree with a first surplus contact between particles a and b . Let f be their youngest separating fork. The independent variables are the actual root positions and velocities of the two fork subtrees and the impact parameters at every creation on their exclusive ancestral paths. Let

$$\mathcal{D}_{ab} : \mathbb{R}^N \longrightarrow \mathbb{R}^3, \quad a \mapsto x_a(t_*) - x_b(t_*)$$

be the derivative at the surplus time. Write $\wedge^2 D\mathcal{D}_{ab}$ for its transverse two-volume.

For a non-grazing collision with flight time τ , the linearized scattering map acts on transverse Jacobi data $(\delta x, \delta v)$ by the standard free-flight/scattering matrices. In curvature coordinates the unstable quadratic form $\mathcal{B}$ satisfies the dispersing Riccati inequality

$$\mathcal{B}^+ \geq \left(\tau + (\mathcal{B}^- + \kappa/\cos \varphi)^{-1} \right)^{-1}.$$

Lemma 1.3 (Flux-weighted exterior Jacobi estimate). *There are $C, p < \infty$ such that for every exclusive path of length m and for every $0 < \delta < 1$,*

$$\int_{\mathcal{E}_m(\delta)} \left| \wedge^2 J_m^{-1} \right|^p d\mathfrak{f}_m \leq C^m \delta^{-C},$$

where J_m is the physical Jacobi map from fork variables to the final transverse relative position, $\mathfrak{f}_m$ is the product incoming-flux measure, and $\mathcal{E}_m(\delta)$ excludes event gaps, relative speeds, and incidence cosines below δ . The complementary flux is at most $C^m \delta^\eta$.

Proof. Use the invariant symplectic Wronskian of two Jacobi fields. Free flight preserves it and a dispersing collision adds the positive second fundamental form divided by the incoming cosine. Hence a two-dimensional fork variation cannot be simultaneously focused without paying inverse powers of an incidence cosine, a short event gap, or a small relative speed. Coarea at one collision integrates each such inverse power against the compensating incoming factor $((v - v_*) \cdot \omega)_+$. Induction along the path gives a factor C per event rather than a deterministic lower singular value. Chebyshev's

inequality yields the bad-flux estimate. No QR rescaling or unit-determinant argument is used. $\square$

Theorem 1.4 (Integrated first-surplus gain). *For a chronological connected cluster with m creation contacts, the absolute contribution of configurations with a first surplus contact is bounded by*

$$\varepsilon^\alpha C^m (1 + T)^m$$

relative to the corresponding creation tree, for some $\alpha > 0$ independent of m . The future after the surplus contact is the true specularly reflected future.

Proof. Condition on all variables except the physical fork variables. On $\mathcal{E}_m(\delta)$, coarea for the two transverse contact equations and Lemma 1.3 give $C^m \varepsilon^2 \delta^{-C}$. The complement has flux $C^m \delta^\eta$. Choose $\delta = \varepsilon^\kappa$ and obtain $\varepsilon^\alpha C^m$. At the first contact apply the actual elastic reflection and restart the positive renewal hierarchy of Proposition 1.2; thus every later collision and mark is retained. The estimate extracts only one geometric gain and pays ordinary C^m factors for the rest. $\square$

Corollary 1.5 (All-depth actual-contact cluster summation). *For sufficiently small kinetic horizon and every bounded complex particle/contact source,*

$$\sum_{m \geq 0} \frac{(CTe^R)^m}{m!} < \infty,$$

and the cyclic sector is $O(\varepsilon^\alpha)$ after all fixed source derivatives. The grand-canonical marked log-Laplace functional is therefore normally convergent on one complex source chart.

Proof. Creation labels, ordered event times, and the factorial hierarchy weight give the displayed exponential series. Insert Theorem 1.4 in the first cyclic block. Cauchy estimates on a slightly larger source ball give the derivative statement. $\square$

1.3. Source-uniform block sewing. Let $\mathcal{S}_{h,\varepsilon}^H$ be the exact hierarchy source propagator on a block of length h , and $\mathcal{S}_h^H$ its Boltzmann tree limit.

Theorem 1.6 (Scaled one-block theorem). *For $0 < h \leq h_0(R)$,*

$$\|\mathcal{S}_{h,\varepsilon}^H - \mathcal{S}_h^H\|_{\alpha,\beta \rightarrow \alpha',\beta'} \leq C_R(h\varepsilon^\alpha + h^2\varepsilon^\eta).$$

The same estimate holds for every fixed source derivative. Consequently, for any partition of $[0, T]$ with mesh h , the accumulated error is

$$C_R T \varepsilon^\alpha + C_R T h \varepsilon^\eta,$$

and has a partition-independent limit as first $\varepsilon \rightarrow 0$ and then $h \rightarrow 0$.

Proof. Expand one block by the compatible renewal series. Noncyclic trees are changed to Boltzmann variables by the ordinary collision coarea formula; the unmatched boundary/end pieces have two event integrations and give $h^2\varepsilon^\eta$. Every cyclic term contains a first surplus and gives $h\varepsilon^\alpha$ by Theorem 1.4.

Normal source convergence permits differentiation term by term. Telescoping the product of contractive block propagators yields the accumulated estimate. The bound is linear, not constant, in h , so it survives refinement. $\square$

1.4. Balance-preserving recovery and the joint LDP. For a smooth positive feasible pair (f, Γ) , subdivide space–time–velocity–angle into finitely many cells. The discrete balance operator is the incidence matrix of the free-transport/contact complex. Its range is the orthogonal complement of the discrete mass, momentum, and energy invariants.

Lemma 1.7 (Conservative finite-cell repair). *Every finite-action feasible pair admits smooth positive approximants (f_n, Γ_n) which satisfy the weak balance exactly and*

$$(f_n, \Gamma_n) \rightarrow (f, \Gamma), \quad \int \ell(d\Gamma_n/dA_{f_n})dA_{f_n} \rightarrow \int \ell(d\Gamma/dA_f)dA_f.$$

Proof. Truncate tails and add a vanishing strictly positive Maxwellian/contact background. Mollify inside each finite cell. The remaining balance defect annihilates the collision invariants. Solve the finite incidence equation on that orthogonal complement and distribute the signed correction over the positive background; its norm tends to zero. Convexity and uniform integrability of perspective entropy give action convergence. No global compact symmetry or energy-preserving Ornstein–Uhlenbeck group is invoked. $\square$

Theorem 1.8 (Grand-canonical and microcanonical joint LDP). *On the weighted density–actual-contact path space, the grand-canonical laws satisfy a good short-time LDP with rate*

$$I_T^{\text{gc}}(f, \Gamma) = I_0^{\text{gc}}(f_0) + \begin{cases} \int_0^T \int \ell\left(\frac{d\Gamma}{dA_f}\right) dA_f dt, & (f, \Gamma) \text{ feasible,} \\ +\infty, & \text{otherwise.} \end{cases}$$

After the B1 exact source-dependent shell conditioning, the same dynamic action holds with the prepared initial rate $I_{0,a}$. Both rates are good.

Proof. Corollary 1.5 and Theorem 1.6 give the full bounded-source pressure and all finite projections. Exponential moments of particle weight, contact mass, and the exact Green balance give compact containment and a Skorokhod modulus. Fenchel duality imposes the closed balance and yields the perspective entropy. For the lower bound, expose a smooth positive pair by finite bounded sources, use the source propagator concentration furnished by the pressure Hessian, and then apply Lemma 1.7. B1 transfers the entire source pressure and the recovery law to the shell with its exact weighted coefficient. Rate sublevels have uniform moment/contact entropy bounds and are therefore compact. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Every cross-paper input is typed in the round-nine dependency ledger. Repository proof and build gates are internal checks and do not replace independent external mathematical review.

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